

Stylized Water Shaders

Requirements

- Tested with Unity 2022.3 LTS and Unity 6.0 LTS.
- Built-in or URP. HDRP is not supported.
- URP usage requires the Universal Render Pipeline package and an active Universal Render Pipeline Asset. The package does not automatically convert a Built-in project to URP.
- A subdivided mesh for vertex waves.
- A camera depth texture for intersection foam.

Quick Start

Paths are relative to `Assets/HaniJahanDesign/Shaders/StylizedWater/`.

1. Open the matching demo scene:
 - Built-in: `Scenes/Shaders_StylizedWater_Demo_BuiltIn.unity`
 - URP: `Scenes/Shaders_StylizedWater_Demo_URP.unity`
2. Assign a preset from `Materials/BuiltIn/` or `Materials/URP/` to a flat, subdivided mesh.
3. Adjust it in the custom material inspector.
4. If using intersection foam, configure camera depth as described below.

To make a material from scratch, select `Hani Jahan Design/StylizedWater/Built-in` or `Hani Jahan Design/StylizedWater/URP` from its **Shader** menu.

Controls

- **Water Surface:** base/shallow height colors, opacity, and edge highlight.
- **Surface Ripples:** procedural Voronoi or a custom texture (red channel), with color, scale, speed, strength, and sharpness controls.
- **Intersection Foam:** a depth-based primary gradient plus optional procedural bands or scrolling texture (red channel).
- **Vertex Waves:** height, scale, and speed. Waves move existing vertices and do not subdivide the mesh.

Negative speed values reverse animation direction.

Camera Depth Setup

Intersection foam requires camera depth. The other water features do not.

Universal Render Pipeline

Enable **Depth Texture** on the active Universal Render Pipeline Asset, or override the water camera's **Rendering > Depth Texture** setting to **On**. When the camera uses the pipeline setting, the active pipeline asset must have **Depth Texture** enabled.

Built-in Render Pipeline

The camera that renders the water may request a depth texture. If needed, add the following component to each applicable camera (or make the equivalent request in existing camera code):

```
using UnityEngine;

[RequireComponent(typeof(Camera))]
public sealed class WaterDepthTexture : MonoBehaviour
{
    private void OnEnable()
    {
        GetComponent<Camera>().depthTextureMode |= DepthTextureMode.Depth;
    }
}
```

Using `|=` preserves depth modes requested by other effects. The shader only reads camera depth while **Intersection Foam** is enabled.

Troubleshooting

- **Pink material:** use the shader matching the active pipeline; for URP, configure the URP package and pipeline asset. Check shader compile errors in the Console.
- **No foam:** enable **Intersection Foam**, configure depth for the Game camera (not only the Scene view camera), and confirm intersecting geometry is inside that camera's culling mask and writes depth. Transparent objects commonly do not write depth.
- **No waves:** enable **Vertex Waves**, increase **Height**, and use a sufficiently subdivided mesh.
- **Incorrect texture patterns:** custom ripple and foam textures use the red channel; check import/wrap settings and adjust scale, speed, and strength.

Known Limitations

- Vertex waves displace existing mesh vertices; they do not add geometry. Low-density meshes can produce angular or barely visible waves.

- Intersection foam depends on the camera depth texture. Transparent objects and other geometry that does not write to depth cannot produce reliable foam intersections.
- The water is transparent, so overlapping transparent surfaces can exhibit Unity's usual transparency sorting artifacts.

Support

Publisher: Hani Jahan Design

Use the support contact on this asset's Unity Asset Store publisher page, or join the [HJD Discord](#) for feedback and community help.